

Examine Bitcoin Price Predictability with Machine Learning Algorithms

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Abstract

As the most-traded digital asset, Bitcoin receives a tremendous increase in investment interests. The primary purpose of this paper is to examine and compare two commonly applied machine learning algorithms on their capability and feasibility in Bitcoin price prediction. The regression models of Extreme Gradient Boosting and Long Short-Term Memory are selected as the investigated objects in this paper. The experiments are evaluated by considering the extra impacts of sample dimensions and time-series frequency, along with the trade-offs between prediction accuracy (measured by residual error) and computational efficiency (measured by computing time). Long Short-Term Memory, as a more convoluted deep learning method, achieves better accuracy when a daily dataset with limited input features is used. However, its predictability has considerably decreased and thus become less efficient than XGBoost, when more peripheral information and higher frequency data points (trading price every 15 minutes) are available. Besides algorithmic complexity, Long Short-Term Memory also takes much longer computing time than Extreme Gradient Boosting, making it a less applicable model to use when dealing with large sample sizes.

Keywords: Bitcoin price prediction, Machine learning algorithms, Fintech, Computing science

1. Introduction

Bitcoin (BTC), the first cryptocurrency, was created during the financial crisis in 2008. It has rapidly gained its marketplace in the international economic landscape by catering to investors' speculative instincts during times of significant market uncertainties and their distrust of financial intermediaries. This investment pattern has been magnified even more since the outbreak of COVID-19, the price of BTC surged by 300% in 2020, but its fluctuation range is often huge, up to ten times higher than that of the fiat currency market [1]. Therefore, it is crucial to predict any insights into the price movements in the BTC market. Compared to other traditional financial assets, cryptocurrencies exhibit two favorable natures, making their market a more suitable test-ground for Artificial Intelligence (AI) algorithmic price prediction experiments [2]. The first is decentralization, meaning they are not issued by the government or central bank and do not need a trusted third party as an intermediary. In contrast, the latter is openness, which refers to its easy market accessibility and transparent transactions. The information on all transactions is public knowledge, and anyone can participate in the trading activities without a central regulating party. The exchange platforms are also open twenty-four-seven without restrictions on trading frequencies.

All these previously mentioned properties of cryptocurrencies would consequently allow Machine Learning (ML) algorithms to learn from their market actions, based on the fact that exchanging prices are driven mainly through behavioral factors and are uncorrelated with other dominant financial instruments [3]. The top-level problem investigated in this paper could be described as forecasting the price of Bitcoin with regressive ML algorithms, which is denoted as the BPR problem for simplicity. Mathematically, the overall regression problem could be represented as the following:

- $\mathbf{X}_g$: matrix of inputs, where $\mathbf{X}_g = [\mathbf{x}_t^v]_g$
- $\mathbf{Y}_g$: vector of labels, where $\mathbf{Y}_g = \{\mathbf{y}_t\}_g$
- L_g : prediction error, where $L_g = \sum_{h \in H} \|y_{c+h,g} - \hat{y}_{c+h,g}\|^2$

The ML models presented in later sections reached the optimised predictions by minimising the L_g function. More specifically,

- t : time index
- v : features index
- g : granularity of Bitcoin price, where $g \in \{5mins, 1440mins\}$

- c : current time
- h : prediction length

The BPR problem has then been separated into two sub-categories, namely BPRD (daily) and BPRM (15-minutes), focusing on different time-series granularity and the numbers of input variables. The following BPRD experiment will test more dimensional but less frequent sample data. While for the BPRM problem, a 15-min interval dataset with only return-related features is applied. This paper compares the model predictability under two circumstances, thus making observations on whether a more granular dataset with a limited number of input variables could compensate for the contribution of various types of features.

The rest of the paper is arranged as follows. The detailed model explanations and formulas are provided in section 2. The preliminary analysis of daily and 15-min datasets, model feature selection, result evaluation matrix, and the model implementation process are all elaborated in section 3. The experimental observations and result discussions are illustrated in section 4. Section 5 presents the conclusion and future elaborations on the current work.

2. Methodology

Two supervised ML algorithms, XGBoost and LSTM, are tested and compared for this research study under both BPRD and BPRM problems. These two particular methods are selected based on their relatively compelling efficacy and accuracy in cryptocurrency price prediction.

2.1. XGBoost

A decision-tree-based ensemble algorithm enables the practical use of Gradient Boosting Decision Trees (GBDTs) by minimizing its tree-growing process [4]. A GBDT, in addition, could be seen as an enhanced version of a decision tree, using base tree models as weak learners. The final tree $\hat{Y}_i^T$ is calculated as:

$$\hat{Y}_i^T = \sum_{k=1}^T f_k(x_i) = \hat{Y}_i^{T-1} + f_T(x_i) \quad (1)$$

where $\hat{Y}_i^{T-1}$ means the tree model generated after T-1 iterations, $f_T(x_i)$ means the base tree model generated in the T-th iteration, and T means the total number of decision tree. Once each tree has been iteratively calculated, the best split could be determined based on the model's scores and gain. The final prediction outcome is achieved by summing the value of each decision tree.

Computational-wise, this method was originally formulated to shorten the time of computing through the allocation of an optimal usage of memory resources. Essential implementation features include dealing with missing values, supporting parallelization in tree construction, fitting and boosting new data added to a trained model [5].

2.2. LSTM

LSTM serves as an improved method based on the traditional Recursive Neural Networks (RNNs), which solves the gradient vanishing problem by adding extra interactions between one memory cell (C_t) and three special gates, namely the input (i_t), forget (f_t) and output gates (o_t) [6]. Each gate output is computed by applying the activation function on the weighted sum of the previous state (S_{t-1}) and the current input (X_t), as expressed in the following formulas:

$$i_t = \sigma(W_i S_{t-1} + W_i X_t) \quad (2)$$

$$f_t = \sigma(W_f S_{t-1} + W_f X_t) \quad (3)$$

$$o_t = \sigma(W_o S_{t-1} + W_o X_t) \quad (4)$$

$$\tilde{i}_t = \sigma(W_{\tilde{i}} S_{t-1} + W_{\tilde{i}} X_t) \quad (5)$$

Then the intermediate cell state ($\tilde{C}_t$), the cell state (C_t) and the hidden state (h_t) are computed as:

$$\tilde{C}_t = \tanh(W_c S_{t-1} + W_c X_t) \quad (6)$$

$$C_t = i_t \odot \tilde{C}_t + f_t \odot C_{t-1} \quad (7)$$

$$h_t = o_t \odot \tanh(C_t) \quad (8)$$

The merits of LSTM are also tested in some pre-recorded experiments. Under the settings of different time horizons, input variables, and parameter specifications, the experiments performed by several previous literature [7-9] all reached the same conclusion that LSTM has outperformed other types of artificial neural networks and produced the most promising forecasts for Bitcoin returns.

More Specifically, there are 3 different types of layers in the model: LSTM Layer, Dropout Layer, and Fully Connected Layer. Among them, LSTM is used to process the time series of BTC price, its output is a vector $(h_1, h_2, \dots, h_n)$, in which n is the number of neurons in the LSTM $h_i (i = 1, 2, 3, \dots, n)$ is the output of the i -th cell in LSTM. Another layer: Dropout Layer, can randomly assign certain parameters the value of 0 to effectively reduce the risk of over fitting. The fully connected layer is to do the linear calculation to reshape the output tensor of LSTM to make its shape meet our requirement. The architecture of the neuron network is showed in Figure 1.

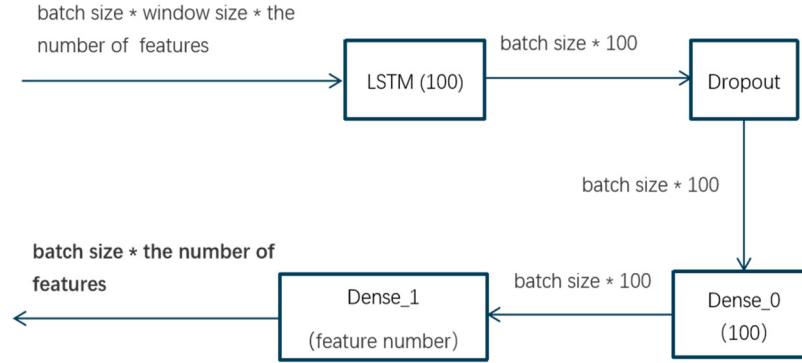


Figure 1. The architecture of Our Neuron Network

3. Experiment

3.1. Dataset Preliminary Analysis

The BPRD dataset includes the aggregated bitcoin daily price, incorporating a complete 4-year's dataset from the 1st day in 2018 to the 1st day in 2022, which was directly downloaded from Yahoo Finance. Figure 2 shows the distribution of the Bitcoin daily closing price. It went through a relatively continuous period from 2018 towards the 3rd quarter of 2020 and has skyrocketed until the beginning of 2021. Then it became highly volatile, surged up, and crashed down in a brief timeframe, and has also kept this momentum in 2022. To consider a complete cycle for Bitcoin price and include some of its volatile nature, a relatively large training set is applied, which composes 85% of the entire dataset.

Table 1. Statistical Properties of BTC daily log-returns.

BPRD (log returns)	Training (1197 obs.)	Test (265 obs.)
Mean (%)	0.1306	-0.088
Median (%)	0.1676	-0.0697
Std (%)	4.0264	4.0375
Skewness	-1.3338	-0.3061
Kurtosis	16.9528	1.2819
Correl (lag1)	-0.0488	-0.0811

Table 2. Statistical Properties of BTC 15-min log-returns.

BPRM (log returns)	Training (13897 obs.)	Test (3480 obs.)
Mean (%)	0.0019	-0.0027
Median (%)	0.0000	-0.0073
Std (%)	0.6092	0.5277
Skewness	0.05975	0.1873
Kurtosis	21.1585	4.9925
Correl (lag1)	-0.00102	0.0248

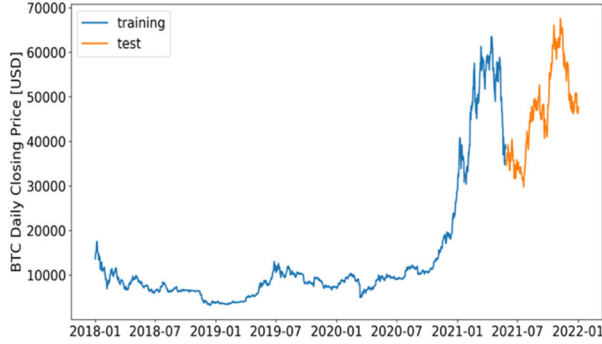


Figure 2. BTC daily price time series plot.

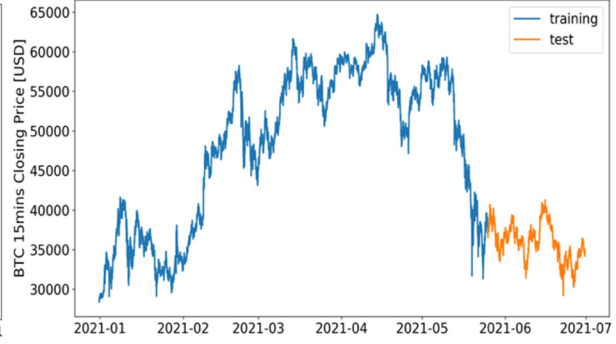


Figure 3. BTC 15-min price time series plot.

Table 1 summarises some statistical characteristics of the log-returns of the daily closing price for both training and testing sets. Notice that Bitcoin experienced positive mean and median returns in training sub-samples but had negative ones in the test set. The relatively high standard deviations in both sub-sets reinforced the idea of the highly volatile nature of cryptocurrencies. The test set has volatility greater than its mean by at least 50. Additionally, the skewness is more than 4-times more damaging in the training set than in the test set. Correl (lag1) indicates the 1st order autocorrelation between the today and the next day's return, which are relatively low in both training and test sets, underlying the fact that a simple linear regression is insufficient to model the relationship between daily returns.

While the BPRM dataset is composed of the 15-min interval Bitcoin real-time trading prices, spanned on a half-yearly basis from January to July 2021. As can be observed from the 4-year time-series plot in Figure 2, a more recent and volatile period is removed in this case in Figure 3. A 5% smaller training set could be used to incorporate its whole cyclical trend. As shown in Table 2, similar statistical properties could also be observed in the 15-min one compared to the daily dataset. In particular, for an even smaller return range, the standard deviation in the test set is nearly 185 times greater than its mean return. In addition, the autocorrelation between 15-mins returns is also relatively small, indicating more input features or complex algorithms are necessary to predict the return for the next period.

3.2. Feature Evaluation

The input feature candidates are selected on top of the analysis of feature importance documented in several related studies, following a predefined categorical framework [10]. The return-predictive independent variables have been grouped into four major classes, namely technical, blockchain, sentiment/interest, and financial instrument-based variables. The detailed information of selected elements and their referenced literature are summarised in Table 3 below. The 'Lag' column shows the number of days (s) that each variable lagged earlier, as the returns of coins are highly interdependent at different time intervals [11]. Notice that the ETH log return is the only feature whose significance has not been verified in previous works. It was also selected due to its nearly perfect correlation ($\rho=0.92$) with the targeted label vector (Y_g).

Table 3. Feature Evaluation.

Feature	Description	Lag	#	Feature type
Log returns	Logarithmic returns of BTC in \$.	1-7	1	Technical
Relative price change	Proxy for periodic volatility, computed with the highest and lowest prices in USD.	1	2	
Parkinson's volatility [12]	Proxy for periodic volatility, computed with the highest and lowest prices in USD.	1-7	3	
Volume [13]	Number of coins exchanged in a day.	1	4	
Gold spot returns [14]	Logarithmic returns of gold spot price in \$. Filled the non-trading days data with its previous day's value.	1-7	5	Instrument-based
ETH log returns	Logarithmic returns of Ethereum in \$.	1-7	6	
Number of transections	The number of daily transactions.	1	7	Blockchain-based [15]
Mining difficulty	A numerical representation of the difficulty on finding a new block.	1	9	
Google trend search volume	The standardised daily search volume of 'Bitcoin'.	1-7	10	Sentiment / Interest-based [16]

3.3. Evaluation Index

These matrices will be used in the later result evaluation session, which are Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE) and R-squared (R2). Their mathematical formulas are also provided below, where y_i , $\bar{y}_i$ and $\hat{y}_i$ are the actual, actual mean and forecasted BTC returns respectively, and N is the number of samples.

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N}} \quad (9)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left(\frac{|y_i - \hat{y}_i|}{y_i} \right) \times 100\% \quad (10)$$

$$R2 = 1 - \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{\sum_{i=1}^N (y_i - \bar{y}_i)^2} \quad (11)$$

Statistically, both RMSE and MAPE are measures of predicted errors. Therefore the smaller the values, the better. The former could be interpreted as the standard deviation of the unexplained variance, and the latter refers to the average deviance between the actual and predicted returns. While the R2 score is an alternative measurement of model accuracy, which can be regarded as the proportion of variability in the actual returns that the regression model could accurately explain, a higher score is desired in this case.

3.4. Model Implementation

With the targeted independent variable being set to the Bitcoin log returns, computed using the closing price in USD. A *Baseline Scenario* is constructed with only BTC price-related technical features (#1-3 in Table 3). The succeeding BPRD experiment will be conducted under the hypothesis that the prediction error (L_g) will decline when other types of features (feature permutations) are added to the baseline scenario. A Best-case Scenario with minimized RMSE value will then be selected and compared. Furthermore, the same set of input features, as in the baseline scenario, will be inserted in the BPRM experiment. This case expects a minor prediction error with more granular sample data. Market efficiency is also assumed under this 15-min data frequency, which means the closing price will be sufficient to reflect all the market information. The corresponding optimized hyperparameter values of XGBoost, and LSTM are models shown in Tables 4 and 5.

Table 4. Optimised hyperparameters for XGBoost.

N_estimators	Max_depth	Gamma	Learning_rate	Min_child_weight	Subsample	Colsample_bytree	Colsample_bylevel
Int = 100	Int = 5	Float = 0	Float = 0.1	Int = 2	Float = 0.9	Float = 1.0	Float = 0.9

Table 5. Optimised hyperparameters for LSTM.

Return sequences	Lstm_neurons	Batchsize	Activation	Loss	Optimiser	Epochs	Dropout
True	100	32	relu	mse	Adam	20	0.2

4. Result and discussion

The values of evaluation indices under the baseline scenario are summarised in Table 6. Computation difficulty, measured by the time used for model implementation, is also included to allow a more practical model comparison.

Table 6. Optimised hyperparameters for LSTM.

	XGBoost	LSTM
RMSE	2323.44	1880.74
MAPE	3.940%	3.215%
R2	0.9439	0.9631
Computational Difficulty	11s	20s

LSTM, in this case, achieves a more accurate prediction, which has a prediction error almost 20% smaller than the XGBoost algorithm. Inconsistent with the previously documented experiment, LSTM is more capable than XGBoost in capturing long-term dependencies from the BPRD dataset across four years. However, computational-wise, it took almost twice the time that XGBoost used to train the regression model for less than 1500 data entries. Model predictability is then improved by enlarging dataset dimensions and increasing datapoint frequency. The enhanced results are presented in Table 7.

Table 7. Optimised hyperparameters for LSTM.

			XGBoost	LSTM
BPRD Scenario	Best-case	RMSE	1628.53	1699.82
		MAPE	2.629%	2.864%
		R2	0.9725	0.9698
		Computational Difficulty	14s	25s
BPRM		RMSE	189.96	174.37
		MAPE	0.387%	0.358%
		R2	0.9937%	0.9931
		Computational Difficulty	118s	304s

The Best-case Scenario for XGBoost is when ETH & Gold prices are added. The RMSE is almost 30% smaller than the Baseline Scenario, while the LSTM method is when Volume is added as an extra variable. The accuracy has only improved by 9.6% in this case. One interesting fact to notice is that when ETH & Gold were used as inputs in the LSTM model, it resulted in a less accurate prediction than the baseline. Similarly, inputting Volume in the XGBoost method only has indiscernible improvement on the benchmark value. Based on the previous observations, XGBoost, as a less complicated machine learning algorithm compared to LSTM, can retrieve more precise predictions when a more significant number of input features are used in the model training process. Notwithstanding, the LSTM model was subjected to the risk of overfitting with a higher dimensional dataset. In addition, LSTM, as customary feed-forward neural networks, is more sensitive to technical features directly related to the regressive object itself. At the same time, XGBoost can tune its regression with exterior information by fitting a new model each time a new predictor is added.

Increasing data frequency also means enlarging sample size under the same time horizon. By altering the granularity from 1440 minutes to 15 minutes, the data points increased by more than ten times in just half the yearly time. Consequently, a much more noticeable improvement occurred when a 10-time larger BPRM dataset was applied for both algorithms, which shrank their corresponding prediction errors more than ten times compared to either the baseline or the best-case scenario. Despite the computation efficiency, LSTM achieves a slightly better result in this case. Even though the sample size increased in the BPRM problem, the prices are retrieved in a smaller scope. Such that the 15-min data only contains 6-month length market information, while the daily one includes information across the entire four years. Therefore, unlike other traditional financial assets, long-term information and its dependencies might not be beneficial in forecasting BTC prices, and the advanced property of the LSTM model could not be fully utilized in this case.

5. Conclusion and future works

In this paper, two machine learning algorithms, XGBoost and LSTM, were employed and compared to predict BTC prices. A baseline scenario was firstly computed with BTC-price-related inputs only. The dataset has been expanded horizontally and vertically by combining various input features and increasing data frequency to improve model predictability. After trying different variable permutations, it could be concluded that XGBoost is more effective than LSTM when training a higher dimensional daily dataset, as it can minimize the prediction error in a much shorter time. More specifically, the XGBoost algorithm is more sensitive to asset-type regressors like gold spot price and Ethereum closing price. LSTM model, on the other hand, could only be improved when BTC price-related features like trading volume are included. It also suggests possible future extensions to this study. For instance, they were using a cryptocurrency portfolio consisting of multiple coins and their derivatives with optimized weight allocations based on their correlations with Bitcoin price as extra input variables. Trying additional macro-economic related indicators like interest rate and hedging commodity prices could be another potential way to improve XGBoost model predictability for the daily dataset.

Increasing data frequency is considered a more effective method to enhance predictive accuracy under both machine learning algorithms. By taking advantage of the unique property of Bitcoin, a more granular closing price can be retrieved from the exchange platform, but this does not apply to other input features. Therefore, instead of a 15-min interval, a smaller or larger granularity, such as 5-min or 30-min ones, could be applied in future analysis to find out the most appropriate time interval for Bitcoin price prediction. There is a trade-off in using LSTM as the regression model in this case, as it achieves a slightly better result than XGBoost by doubling the computing time. Hence, XGBoost might be a more suitable algorithm when high-frequency data is available in a larger sample size. Furthermore, in light of the feature adjustability of XGBoost, the previously suggested cryptocurrency portfolio could also be applied with less concern for the risk of overfitting.

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